RMMM PLAN AND RISK INFORMATION SHEETS

Theory:

Risk and Risk Management: A risk is an uncertain event or condition that, if it occurs, negatively affects the cost, schedule, quality or success of a project. Every risk has two characteristics: uncertainty (it may or may not happen) and loss (there is an unwanted consequence if it does). Risk management is the process of identifying risks, estimating them and planning actions to control them so that the project can be completed successfully.

RMMM: RMMM stands for Risk Mitigation, Monitoring and Management. It is a plan that documents how each identified risk will be handled:

Risk Mitigation – steps taken in advance to reduce the probability of the risk occurring or to reduce its impact.

Risk Monitoring – tracking the project to check whether the risk is becoming more likely, whether the mitigation steps are working, and to collect information for future risk analysis.

Risk Management – the contingency plan: actions carried out after the risk has actually occurred, started when a defined trigger condition is met.

Risk Projection: Each identified risk is rated in two ways. Probability (P) is the likelihood that the risk will occur. Impact is the seriousness of the consequences if it occurs, classified into four categories: 1 – Catastrophic, 2 – Critical, 3 – Marginal and 4 – Negligible.

Risk Exposure (RE): Risk exposure combines likelihood and loss into one number: RE = P × C, where P is the probability of the risk occurring and C is the estimated cost to the project if it occurs. Risks are ranked by RE so that the team spends its effort on the highest-exposure risks first, and prepares detailed Risk Information Sheets for them.

Estimated Cost (C) and Impact: The estimated cost of a risk grows with the seriousness of its impact, i.e. Catastrophic > Critical > Marginal > Negligible. For the E-Waste Management System, the cost ranges were fixed as a share of the estimated project cost of about Rs. 21,00,000, as shown below. Each risk's cost lies within the range of its impact category and reflects rework, recovery, delay and support effort for this project.

Impact Category	Effect on the E-Waste Management System	Estimated Cost (C) Range
Catastrophic (1)	Project failure or severe damage — e.g. confidential corporate data or Certificates of Destruction compromised, system unusable.	Rs. 4,00,000 and above (about 20% or more of project cost)
Critical (2)	Major functionality impaired — significant rework, delay or loss of service quality.	Rs. 1,00,000 to Rs. 3,00,000 (about 5–15%)
Marginal (3)	Minor functionality or performance affected — manageable rework.	Rs. 20,000 to Rs. 60,000 (about 1–3%)
Negligible (4)	Small inconvenience — no real effect on cost or schedule.	Below Rs. 20,000 (under 1%)

Risk Information Sheet (RIS): A Risk Information Sheet records one risk in a standard format — date, probability, impact, risk ID, description, refinement/context (sub-conditions), mitigation/monitoring steps, management (contingency plan and RE), current status, originator and the person or team assigned. In this experiment 8 risks (R1–R8) were identified for the

E-Waste Management System, and Risk Information Sheets are prepared for the six with the highest risk exposure (R1–R6); R7 and R8 have low exposure and are only tracked in the table.

Identifying Risks:

Risk ID	Risk Summary	Probability	Impact	Estimated Cost (C)	Risk Exposure (RE = P × C)
R1	Unauthorized Access to Confidential Corporate Data and Certificates	40%	Catastrophic (1)	Rs. 5,00,000	Rs. 2,00,000
R2	Live Metal Price Feed Failure or Delay	70%	Critical (2)	Rs. 2,00,000	Rs. 1,40,000
R3	Server, PostgreSQL or Redis Failure	30%	Critical (2)	Rs. 3,00,000	Rs. 90,000
R4	OpenStreetMap Routing Failure or Inaccurate Routes	60%	Critical (2)	Rs. 1,50,000	Rs. 90,000
R5	Certificate of Destruction Not Generated or Delivered Correctly	40%	Critical (2)	Rs. 1,50,000	Rs. 60,000
R6	Driver Packing-Verification Photo Upload Failure	55%	Critical (2)	Rs. 1,00,000	Rs. 55,000
R7	Damaged or Unreadable QR/Barcode Labels and Scanner Issues	50%	Marginal (3)	Rs. 40,000	Rs. 20,000
R8	Minor UI Inconsistencies Across Browsers and Devices	25%	Negligible (4)	Rs. 10,000	Rs. 2,500

Impact:
1 - Catastrophic
2 - Critical
3 - Marginal
4 - Negligible

Risk Information Sheets:

Risk Information Sheet: R1 (Unauthorized Access to Confidential Corporate Data and Certificates)

Field	Details
Date	11/10/2026
Probability	40%
Impact	Catastrophic
Risk ID	R1
Description	Unauthorized users may gain access to confidential data such as corporate asset records, data-wipe requirements, chain-of-custody logs and Certificates of Destruction, leading to privacy violations, loss of compliance credibility and misuse of data.
Refinement/Context	**Sub condition 1:** Weak passwords or poor authentication may allow attackers to log in to user accounts. **Sub condition 2:** Role-based access control may be improperly configured, allowing users (for example, non-'Recycler_Admin' users) to access features or data beyond their role. **Sub condition 3:** Unencrypted data transmission or common attacks such as SQL injection may expose asset tracking and corporate records.
Mitigation/Monitoring	1. Enforce Role-Based Access Control and authenticate users before granting access to protected features such as certificate issuing. 2. Store passwords in hashed form, transmit all data over TLS 1.2+ and encrypt corporate records at rest. 3. Validate all user inputs to prevent SQL injection and other common security attacks. 4. Monitor failed login attempts, suspicious access patterns and unauthorized access attempts.
Management	RE computed to be Rs. 2,00,000 (Probability 40% × Estimated Cost Rs. 5,00,000, based on data breach handling, security audit, legal and recovery effort).
Current Status	Mitigation steps to be initiated.
Originator	E-Waste Management System Project Team
Assigned	Development Team and System Administrator

Risk Information Sheet: R2 (Live Metal Price Feed Failure or Delay)

Field	Details
Date	11/10/2026
Probability	70%
Impact	Critical
Risk ID	R2
Description	The third-party metal market API may be unavailable, slow or return invalid data, so the Smart Valuation Module cannot fetch current copper, aluminum and gold rates and users receive no quote or an outdated, unfair quote.
Refinement/Context	**Sub condition 1:** The external metal price API may be down, rate-limited or may change its response format. **Sub condition 2:** Slow responses may exceed the 2-second response target for real-time price queries. **Sub condition 3:** Stale or incorrect prices may be shown to users if cached values in Redis are not refreshed or flagged.
Mitigation/Monitoring	1. Cache the latest valid prices in Redis and use them as a fallback, clearly labelled with their last-updated time. 2. Validate API responses (range and format checks) before using them for quote calculation. 3. Use timeouts, retries and, where possible, a secondary price data source. 4. Monitor API uptime, response times, failed fetches and the age of cached prices.
Management	RE computed to be Rs. 1,40,000 (Probability 70% × Estimated Cost Rs. 2,00,000, based on integration fixes, fallback development, incorrect quote correction and user support effort).
Current Status	Mitigation steps to be initiated.
Originator	E-Waste Management System Project Team
Assigned	Development Team

Risk Information Sheet: R3 (Server, PostgreSQL or Redis Failure)

Field	Details
Date	11/10/2026
Probability	30%
Impact	Critical
Risk ID	R3
Description	The Node.js application server, PostgreSQL database or Redis cache may fail, making the system unavailable and preventing users from logging e-waste, drivers from viewing schedules and recyclers from updating asset status.
Refinement/Context	**Sub condition 1:** Hardware failure, crash or overload of the containerized application server may bring the system down. **Sub condition 2:** Database corruption or connection-pool exhaustion may make asset and tracking records inaccessible. **Sub condition 3:** Absence of recent backups may cause permanent loss of chain-of-custody data and prolonged downtime.
Mitigation/Monitoring	1. Perform regular PostgreSQL backups and periodically test data restoration. 2. Configure a standby database and container auto-restart for quick recovery. 3. Use health checks on Docker containers, with automatic restart of failed services. 4. Monitor server uptime, database connections, Redis memory, disk usage and error logs.
Management	RE computed to be Rs. 90,000 (Probability 30% × Estimated Cost Rs. 3,00,000, based on server and database recovery, data restoration and downtime losses).
Current Status	Mitigation steps to be initiated.
Originator	E-Waste Management System Project Team
Assigned	System Administrator

Risk Information Sheet: R4 (OpenStreetMap Routing Failure or Inaccurate Routes)

Field	Details
Date	11/10/2026
Probability	60%
Impact	Critical
Risk ID	R4
Description	The OpenStreetMap/Leaflet routing service may be unavailable or return inaccurate or slow routes, so drivers receive poor pickup routes, fuel costs rise and hazardous-material transport risk increases.
Refinement/Context	**Sub condition 1:** The free OpenStreetMap service may be unavailable or rate-limited during peak usage. **Sub condition 2:** Route calculation may take longer than the 5-second target for up to 20 daily stops. **Sub condition 3:** Incorrect addresses or map data may lead drivers to wrong pickup locations.
Mitigation/Monitoring	1. Cache computed routes and show the unoptimized pickup list as a fallback when routing fails. 2. Validate and geocode pickup addresses at the time of asset submission. 3. Limit route requests and consider a self-hosted routing instance if usage grows. 4. Monitor routing API failures, calculation times and driver-reported route problems.
Management	RE computed to be Rs. 90,000 (Probability 60% × Estimated Cost Rs. 1,50,000, based on routing fallback development, schedule rework and delayed pickups).
Current Status	Mitigation steps to be initiated.
Originator	E-Waste Management System Project Team
Assigned	Development Team

Risk Information Sheet: R5 (Certificate of Destruction Not Generated or Delivered Correctly)

Field	Details
Date	11/10/2026
Probability	40%
Impact	Critical
Risk ID	R5
Description	The Certificate of Destruction may fail to generate, be generated with incorrect data, or not be emailed to the corporate client after data drives are destroyed, causing loss of compliance evidence and client trust.
Refinement/Context	**Sub condition 1:** Certificate generation may be triggered before FR-02 (wipe requirement) and FR-09 (breakdown time log) are properly completed. **Sub condition 2:** Email delivery may fail due to an invalid address or mail server issues. **Sub condition 3:** Signing or verification of the certificate may fail, making it not cryptographically verifiable.
Mitigation/Monitoring	1. Generate a certificate only when wipe requirement and destruction time records are complete and the user holds the 'Recycler_Admin' role. 2. Use a retry queue for email delivery and let clients re-download certificates from their dashboard. 3. Apply cryptographic signing and test certificate verification regularly. 4. Monitor failed certificate generations, undelivered emails and verification errors.
Management	RE computed to be Rs. 60,000 (Probability 40% × Estimated Cost Rs. 1,50,000, based on certificate re-issue, signing fixes and client support effort).
Current Status	Mitigation steps to be initiated.
Originator	E-Waste Management System Project Team
Assigned	Development and Testing Team

Risk Information Sheet: R6 (Driver Packing-Verification Photo Upload Failure)

Field	Details
Date	11/10/2026
Probability	55%
Impact	Critical
Risk ID	R6
Description	Drivers may be unable to upload the packing-verification photo (FR-07) due to poor network, device or camera problems, blocking pickups or tempting drivers to skip the safety check for bloated batteries and exposed toxics.
Refinement/Context	**Sub condition 1:** Poor mobile network at pickup locations may cause uploads to fail or time out. **Sub condition 2:** Large photo files from high-resolution cameras may exceed upload limits or be rejected. **Sub condition 3:** Frequent failures may delay pickups and discourage drivers from following the safety workflow.
Mitigation/Monitoring	1. Save the photo locally and retry synchronization automatically when the network returns. 2. Compress or resize photos before upload and show clear error messages. 3. Keep the photo requirement enforced; allow only a dispatcher-approved override with a recorded reason. 4. Monitor upload failure rate, retry counts and pickups pending verification.
Management	RE computed to be Rs. 55,000 (Probability 55% × Estimated Cost Rs. 1,00,000, based on offline sync development, delayed pickups and safety-process rework).
Current Status	Mitigation steps to be initiated.
Originator	E-Waste Management System Project Team
Assigned	Development Team